

Text Preprocessing

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آکادمی‌ها



Variations of Corpora

- Language: 7097 languages in the world
- Variety, like African American Language varieties.
 - ▶ AAE Twitter posts might include forms like "iont" (I don't)
- Code switching, e.g., Spanish/English, Hindi/English:

S/E: Por primera vez veo a @username actually being hateful! It was beautiful:)

[For the first time I get to see @username actually being hateful! it was beautiful:)]

H/E: dost tha or ra- hega ... dont worry ... but dherya rakhe

["he was and will remain a friend ... don't worry ... but have faith"]
- Genre: newswire, fiction, scientific articles, Wikipedia
- Author Demographics: writer's age, gender, ethnicity, SES



➤ **Tokenization**

➤ **Normalization**

➤ **Lemmatization**

➤ **Stemming**

➤ **Stopword Removal**



Space-based Tokenization

- A very simple way to tokenize
 - ▶ For languages that use space characters between words
 - ▶ Arabic, Cyrillic, Greek, Latin, etc., based writing systems
 - ▶ Segment off a token between instances of spaces
- Unix tools for space-based tokenization
 - ▶ The "tr" command
 - ▶ Inspired by Ken Church's UNIX for Poets
 - ▶ Given a text file, output the word tokens and their frequencies



More Counting

➤ Merging upper and lower case

```
tr 'A-Z' 'a-z' < shakes.txt | tr -sc 'A-Za-z' '\n' | sort | uniq -c
```

➤ Sorting the counts

```
tr 'A-Z' 'a-z' < shakes.txt | tr -sc 'A-Za-z' '\n' | sort | uniq -c | sort -n -r
```

```
23243 the
22225 i
18618 and
16339 to
15687 of
12780 a
12163 you
10839 my
10005 in
8954 d
```

What happened here?



Issues in Tokenization

➤ Can't just blindly remove punctuation:

- ▶ m.p.h., Ph.D., AT&T, cap'n
- ▶ prices (\$45.55)
- ▶ dates (01/02/06)
- ▶ URLs (<http://www.stanford.edu>)
- ▶ hashtags ([#nlproc](#))
- ▶ email addresses (someone@cs.colorado.edu)

➤ Clitic: a word that doesn't stand on its own

- ▶ "are" in [we're](#), French "je" in [j'ai](#), "le" in [l'honneur](#)

➤ When should multiword expressions (MWE) be words?

- ▶ [New York](#), [rock 'n' roll](#)



Tokenization in NLTK

Bird, Loper and Klein (2009), Natural Language Processing with Python. O'Reilly

```
>>> text = 'That U.S.A. poster-print costs $12.40...'
>>> pattern = r'''(?x)      # set flag to allow verbose regexps
...     ([A-Z]\.)+          # abbreviations, e.g. U.S.A.
...     | \w+(-\w+)*        # words with optional internal hyphens
...     | \$?\d+(\.\d+)?%?   # currency and percentages, e.g. $12.40, 82%
...     | \.\.\.            # ellipsis
...     | [][.,;"'()?)-_']  # these are separate tokens; includes ], [
... '''
>>> nltk.regexp_tokenize(text, pattern)
['That', 'U.S.A.', 'poster-print', 'costs', '$12.40', '...']
```



Tokenization in Languages without Spaces

- Many languages (like Chinese, Japanese, Thai) don't use spaces to separate words!
- How do we decide where the token boundaries should be?



How to do word tokenization in Chinese?

姚明进入总决赛 "Yao Ming reaches the finals"



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➤ 3 words?

姚明 进入 总决赛

YaoMing reaches finals



How to do word tokenization in Chinese?

姚明进入总决赛 "Yao Ming reaches the finals"

➤ 3 words?

姚明 进入 总决赛

YaoMing reaches finals

➤ 5 words?

姚 明 进入 总 决赛

Yao Ming reaches overall finals

➤ 7 characters? (don't use words at all):

姚 明 进 入 总 决 赛

Yao Ming enter enter overall decision game



Sentence Segmentation

- !, ? mostly unambiguous but period "." is very ambiguous
 - ▶ Sentence boundary
 - ▶ Abbreviations like Inc. or Dr.
 - ▶ Numbers like .02% or 4.3
- Common algorithm: Tokenize first: use rules or ML to classify a period as either (a) part of the word or (b) a sentence-boundary.
 - ▶ An abbreviation dictionary can help
- Sentence segmentation can then often be done by rules based on this tokenization.



- Tokenization
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Word Normalization

➤ Putting words/tokens in a standard format

▶ U.S.A. or USA

▶ uhhuh or uh-huh

▶ Fed or fed

▶ am, is, be, are



➤ Applications like IR: reduce all letters to lower case

- ▶ Since users tend to use lower case
- ▶ Possible exception: upper case in mid-sentence?
 - ▶ e.g., **General Motors**
 - ▶ **Fed** vs. **fed**
 - ▶ **SAIL** vs. **sail**

➤ For sentiment analysis, MT, Information extraction

- ▶ Case is helpful (**US** versus **us** is important)



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Lemmatization

➤ Represent all words as their lemma, their shared root
= dictionary headword form:

▶ am, are, is → be

▶ car, cars, car's, cars' → car

▶ Spanish **quiero** ('I want'), **quieres** ('you want')

→ **querer** 'want'

▶ He is reading detective stories

→ He be read detective story



Lemmatization is done
by Morphological Parsing

➤ Morphemes:

- ▶ The small meaningful units that make up words
- ▶ **Stems:** The core meaning-bearing units
- ▶ **Affixes:** Parts that adhere to stems, often with grammatical functions

➤ Morphological Parsers:

- ▶ Parse cats into two morphemes cat and s
- ▶ Parse Spanish amaren ('if in the future they would love') into morpheme amar 'to love', and the morphological features 3PL and future subjunctive.



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Stemming

➤ Reduce terms to stems, chopping off affixes crudely

This was not the map we found in Billy Bones's chest, but an accurate copy, complete in all things-names and heights and soundings-with the single exception of the red crosses and the written notes.



Thi wa not the map we found in Billi Bone s chest but an accur copi complet in all thing name and height and sound with the singl except of the red cross and the written note.



Porter Stemmer

- Based on a series of rewrite rules run in series
 - ▶ A cascade, in which output of each pass fed to next pass
- Some sample rules:

ATIONAL → ATE (e.g., relational → relate)

ING → ε if stem contains vowel (e.g., motoring → motor)

SSES → SS (e.g., grasses → grass)



Dealing with complex morphology

➤ Dealing with complex morphology is necessary for many languages
e.g., the Turkish word:

➤ **Uygarlastiramadiklarimizdanmissinizcasina**

➤ '(behaving) as if you are among those whom we could not civilize'

➤ **Uygar** 'civilized' + **las** 'become'

➤ + **tir** 'cause' + **ama** 'not able'

➤ + **dik** 'past' + **lar** 'plural'

➤ + **imiz** 'p1pl' + **dan** 'abl'

➤ + **mis** 'past' + **siniz** '2pl' + **casina** 'as if'



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- Many of the most frequently used words in English are likely to be useless for text mining
- These words are called Stopwords
 - ▶ Examples: the, of, and, to, an, is, that, ...
 - ▶ Typically text contains about 400 to 500 such words
 - ▶ For an application, an additional domain specific stopwords list may be constructed.



Stopword Removal

➤ Why should we remove stopwords?

- ▶ Reduce data set size
- ▶ Stopwords account for 20-30% of total word count
- ▶ Improve effectivity of text mining methods
- ▶ Stopwords may confuse the mining algorithm

➤ Challenges

- ▶ to be or not to be



Further Reading

► Speech and Language Processing (3rd ed. draft)

► Chapter 2